A-A-51129B
May 21, 1992
SUPERSEDING
A-A-51129A
September 2, 1991

COMMERCIAL ITEM DESCRIPTION

MANDRELS, MACHINE, SOLID

The General Services Administration has authorized the use of this commercial item description as a replacement for A-A-51129A, type I.

1. ABSTRACT

1.1 Abstract. This description covers solid mandrels which are used for supporting workpieces between centers for machining operations.

2. CLASSIFICATION

2.1 Classification. The mandrels shall be of the sizes listed in table 1. The size furnished shall be as specified (see table 1 and 7.1).

3. SALIENT CHARACTERISTICS

3.1 Design. The mandrels shall be cylindrically shaped with a taper of 0.0005 inch +10 percent per inch on the overall length. Each end of the mandrel shall be centered and have a milled flat for a holding device. The mandrels shall be similar to figure 1.

3.2 Hardness. The mandrels shall have a hardness of not less than 58 and not more than 65 on the Rockwell "C" scale when measured in accordance with ASTM E 18.

3.3 Tolerance and finish. The mandrel centers shall be machined to a standard angle of 60 degrees ±30 minutes. The surface roughness average (Ra) of the mandrels shall be not greater than 20 microinches using a 0.030 inch cutoff, when measured in accordance with ANSI B46.1.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense General Supply Center, ATTN: DGSC-SS, 8000 Jefferson Davis Highway, Richmond, VA 23297-5100.

AMSC N/A FSC 3460

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3.4 Identification marking. Each mandrel furnished shall be permanently marked with the size or the manufacturer's part number and the manufacturer's name or trademark of such known character that the source of the manufacturer may be identified.

TABLE I. Sizes and dimensions.

Size	Diameter 1/	Length 2/	Size	Diameter 1/	Length 2/	Size	Diameter 1/	Length 2/
1	1/8	2-1/2	19	7/8	6-1/2	37	2	11
2	5/32	3	20	15/16	6-3/4	38	2-1/8	11-1/2
3	3/16	3	21	1	7	39	2-1/4	12
4	7/32	3-3/4	22	1-1/16	7-1/4	40	2-3/8	12
5	1/4	3-3/4	23	1-1/8	7-1/2	41	2-1/2	12-1/2
6	9/32	4	24	1-3/16	7-3/4	42	2-5/8	12-1/2
7	5/16	4	25	1-1/4	8	43	2-3/4	13
8	11/32	4-1/4	26	1-5/16	8-1/4	44	2-7/8	13
9	3/8	4-1/4	27	1-3/8	8-1/2	45	3	13
10	13/32	4-1/2	28	1-7/16	8-3/4	46	3-1/4	13-1/2
11	7/16	4-1/2	29	1-1/2	9	47	3-1/2	14
12	15/32	5	30	1-9/16	9-1/4	48	3-3/4	14-1/2
13	1/2	5	31	1-5/8	9-1/2	49	4	15
14	9/16	5-1/4	32	1-11/16	9-3/4			
15	5/8	5-1/2	33	1-3/4	10			
16	11/16	5-3/4	34	1-13/16	10-1/4			
17	3/4	6	35	1-7/8	10-1/2			
18	13/16	6-1/4	36	1-15/16	10-3/4			

NOTE: All dimensions are in inches.

1/ The diameter of the mandrel shall be measured at a point between L/3 and L/2, where L is the mandrel length.

2/ Tolerance of length is ± 1/2 inch.

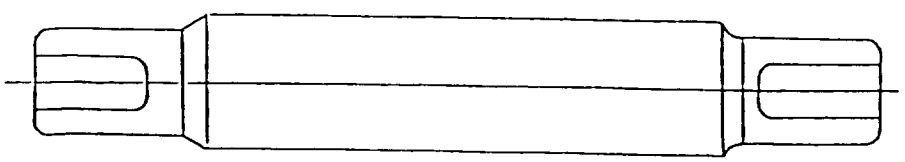


FIGURE 1. Mandrel, solid type.

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3.5 General requirements.

3.5.1 Environmental protection. During the manufacture of the mandrels, materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect on the date of the contract shall not be used.

3.5.2 Reference documents. The documents referenced in this document shall be the issues in effect on the date of issuance of the invitation for bid or request for proposal.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent possible.

5. QUALITY ASSURANCE PROVISIONS

5.1. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PRESERVATION, PACKING, LABELING, AND MARKING

6.1 Preservation, packing, labeling, and marking. Preservation, packing, labeling, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Size, and quantity required.

7.2 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

AA51129 XX
- XX: Size (see table I)
- AA51129: CID number

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7.3 Source for documents referenced in this CID.

ANSI B46.1 is available from the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.

ASTM E 18 is available from American Society for testing and materials, 1916 Race Street, Philadelphia, PA 19103-1187.

MILITARY INTEREST:

Custodians

Army - AL
Navy - YD
Air Force - 99

Review Activities

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CIVIL AGENCY COORDINATION ACTIVITY:

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PREPARING ACTIVITY:

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Project 3460-0169

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-51129B	92 May 21

3. DOCUMENT TITLE
MANDRELS, MACHINE, SOLID

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	(1) Commercial (901)775-4749	(2) AUTOVON 683-4749
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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